

36/50

ABERDARE TECHNICAL COLLEGE

**UNIT OF COMPETENCY: DESIGN COMPUTERIZED
DATABASE**

LEVEL: 5

UNIT CODE: 0612 451 08A

ELEMENT COVERED: DESIGN COMPUTERIZED DATABASE

CAT 1, TERM 3

TIME: 1 HOUR 30 MINUTES

TOTAL MARKS: 50

Instructions to Candidates

- ✓ *Write your name and admission number in the space provided.*
- ✓ *This paper has **THREE** sections: A, B, and C. Answer **ALL** questions in all sections.*
- ✓ *Write your answers legibly in the spaces/answer booklet provided.*
- ✓ *Do not write on the question paper.*

Name: Brian Mquina Admission no.....Date:

~~14~~ 14/09/2026 sign. [Signature]

① Database → Organized Collection of data for easier access ✓ 8

② Database Management System → a ^{software} system used to create store and retrieve data. 8

③ Data redundancy → Replication of data ✓ 8

④ Data independence → ability of data to be on its own without depending on other 4

⑤ Functions of Database Management System.

Data Storage ✓

Data Sharing ✓ 8

Data Retrieval ✓

Data insertion ✓

3 Application of database

Hospital → keep Patients' data. ①

Business → Calculate Loss and Profit. 1/2 8

Government → keep Records. 1/2

Banking → keep Records which Records? 1/2

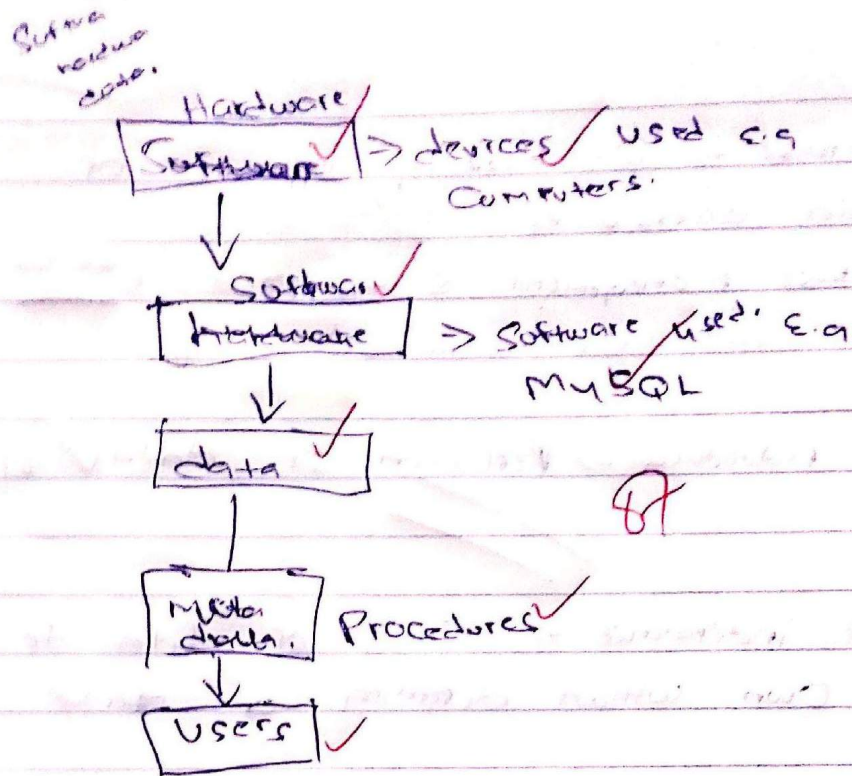
① Centralized → data is stored and managed in one central position. ✓

Distributed → data is stored across multiple servers but they work together ✓

② Easier to manage and maintain. 8
Cost Effective 4

4

⑥



⑦

Explain the 3-tier database Architecture.

They have 3 tier.

Presentation tier → User interface → e.g. Student.

Application tier → Apply Rules of Application. ~~Completed~~

Data tier. → DBMS is Stored. ~~Database~~

DB

data + DBMS

⑧ Requirement Elicitation. → The process of Researching and discovering the requirement of design and drawing from user. ✓

⑨ Questionnaire ✓
Interviews ✓

⑩ 3 Components that should be included in a database Assessment Report.

1. Introduction. ✓

2. ~~System~~ User Requirements ✓

3. ~~Backup~~ Current System Requirements. ✓

⑪ Structure of data: Presentation

Hierarchical data is presented in a tree like structure while in Relational data is presented in tables. ✓

ii Relationships between Entities/records.
Relational Supports Relationship Such as One to many. Many to many using keys while hierarchical Supports. one to many. ~~OL~~ OL

iii Flexibility in querying data
Relational is very flexible while in hierarchical is less flexible. OL

iv ~~Relationship~~ Employee to Employees

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